# How Cardiovascular Conditions Affect Heart Disease

**Group number 05**

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# 1. Acknowledgement

This is to express our warmest gratitude to Dr. Dilhari Attygalle for nourishing us with knowledge and guiding us through this project. And we would like to extend our thanks to the respective Assistant lecturers for providing guidance and pointing out any short comings. Our gratitude goes to the University of Colombo for providing this valuable opportunity. Last but not least we are thankful to our parents and friends for supporting us throughout the project directly and indirectly.

# 2. Executive Summary

This project analyzes the cardiovascular data set and aims to enhance understanding and management of cardiovascular health. By identifying key risk factors associated with heart disease, the analysis aims to build predictive models that can accurately forecast the likelihood of heart disease based on patient data.

- Key findings
  - Age as a risk factor
  - Gender disparity
  - Chest Pain as an Indicator
  - Blood Pressure, Cholesterol Levels, Diabetes Link, Exercise-Induced Angina as factors
  - ECG Abnormalities and maximum heart rate as analysts
  - number of blocked major vessels as resources for classification of heart diseases

- Recommendations
  - Routine Screening
  - Hypertension Management
  - Cholesterol Control and Diabetes Monitoring
  - Symptom Awareness
  - ST Segment Monitoring
  - Comprehensive Vessel Assessments
  - Preventive Healthcare

# 3. Introduction

**I.** Cardiovascular conditions encompass a range of disorders affecting the heart and blood vessels, and they are a leading cause of heart disease. These conditions include hypertension (high blood pressure), atherosclerosis (narrowing of the arteries due to plaque buildup), arrhythmias (irregular heartbeats), heart valve disorders, and heart failure.Several factors contribute to the development of cardiovascular conditions, many of which significantly increase the risk of heart disease. Some of the key factors include high blood pressure, diabetes, high cholesterol levels and exercise induced angina.

High blood pressure puts extra strain on the heart and blood vessels, leading to heart muscle damage over time. Unhealthy diets high in cholesterol, sugar, and fats can lead to conditions like diabetes and high

cholesterol, which contribute to plaque buildup in the arteries (atherosclerosis). This plaque narrows and hardens the arteries, restricting blood flow to the heart and increasing the risk of coronary artery disease. If left untreated, these conditions can lead to various forms of heart disease, such as coronary artery disease, heart failure, or even sudden cardiac arrest. Early detection and management of cardiovascular conditions are crucial in reducing the risk of heart disease and maintaining overall heart health.

The heart disease dataset in this report mentioned originated from a multispecialty hospital in India and **comprises data from 946 subjects**. It includes **13 features**, which are commonly associated with heart disease. This dataset is particularly valuable for researchers and data scientists aiming to develop early-stage heart disease detection systems and create predictive models. Given the growing incidence of heart disease globally, especially in countries like India, this dataset provides a significant opportunity to improve diagnostic accuracy and contribute to preventive healthcare measures.

# ii. About the data set

Datasource:https://www.kaggle.com/datasets/jocelyndumlao/cardiovascular-disease-dataset

# iii. Introduction of variables

| S.No | Attribute | Assigned Code | Unit | Type of the Data |
|---|---|---|---|---|
| 1 | Patient Identification Number | patientid | Number | |
| 2 | Age | age | In Years | Quantitative |
| 3 | Gender | gender | 1,0(0= female, 1 = male) | Qualitative |
| 4 | Chest pain type | chestpain | 0,1,2,3 (Value 0: typical angina Value 1: atypical angina Value 2: non-anginal pain Value 3: asymptomatic) | Qualitative |
| 5 | Resting blood pressure | restingBP | 94-200 (in mm HG) | Quantitative |
| 6 | Serum cholesterol | serumcholestrol | 126-564 ( in mg/dl) | Quantitative |
| 7 | Fasting blood sugar | fastingbloodsugar | 0,1 > 120 mg/dl (0 = false , 1 = true) | Qualitative |
| 8 | Resting electrocardiogram results | restingrelectro | 0,1,2 (Value 0: normal, Value 1: having ST-T wave abnormality (T wave inversions and/or ST elevation or depression of > 0.05 mV), Value 2: showing probable or definite left ventricular hypertrophy by Estes' criteria) | Qualitative |
| 9 | Maximum heart rate achieved | maxheartrate | 71-202 | Quantitative |
| 10 | Exercise induced angina | exerciseangia | 0,1 (0 = no, 1 = yes) | Qualitative |
| 11 | Oldpeak =ST | oldpeak | 0-6.2 | Quantitative |
| 12 | Slope of the peak exercise ST segment | slope | 1,2,3 (1-upsloping, 2-flat, 3-downsloping) | Qualitative |
| 13 | Number of major vessels | noofmajorvessels | 0,1,2,3 | Quantitative |
| 14 | Classification of heart disease | target | 0,1 (0= Absence of Heart Disease, 1= Presence of Heart Disease) | Qualitative |

## Chest pain types:

0=Typical Angina- when the heart muscle doesn't get enough oxygen-rich blood it might cause chest pain which typically happens with physical exercise or stress**. Here its easier to detect the reason**

1=Atypical angina- Atypical angina may have less obvious symptoms or occur under different conditions. Atypical angina is more commonly seen in women and elderly people. **Detecting this is also somewhat easy**

2=Non-angina pain does not originate from reduced blood flow to the heart muscle. It can have various causes, often unrelated to the cardiovascular system. **Detecting the root can be relatively hard**

3=Asymptomatic - refers to the absence of symptoms despite the presence of a condition or disease. They do not exhibit any noticeable signs or symptoms of illness, **detecting the root is considerably hard.**

## Resting electrocardiogram results

**0** - normal readings(reading shown is in the acceptable region)
**1** – readings that correspond to ST-T wave abnormalities being present. Which may be indicative of the chances of getting heart diseases such as heart attacks
**2** – readings that indicate the chances of left ventricle thickening being high. Which increases the risk of getting heart diseases.

## Exercise-induced angina

Exercise-induced angina is chest pain that happens when your heart doesn't get enough blood during exercise or physical activity. It's often a sign of heart disease. Common symptoms include chest tightness, pain in your arms or neck, and trouble breathing. Things like

exercise, stress, or cold weather can bring it on. Treatment includes healthy lifestyle changes, medications, and sometimes surgery to improve blood flow

The following three conditions deal with characteristics of blood vessels

**Slope**: "slope" refers to curvature of the blood vessel.
**Up sloping**: refers to upward bend of blood vessel - Generally normal.
**Flat** - refers to blood vessel being straight - May indicate mild ischemia (reduced blood flow).

**Down sloping** - refers to downward bend of blood vessels - Indicates significant ischemia and higher heart disease risk. These patterns help assess the likelihood of coronary artery disease

**Old Peak**: "oldpeak" refers to a measurement obtained from the ECG wave. high “oldpeak" values mean there is more significant restriction in blood flow to the heart. Lower "oldpeak" values suggest less restriction. It is measured in millimeters (mm). A higher reading suggests a higher risk of heart disease.

### The Number of Major Vessels

Refers to the number of blood vessels blocked
**0 vessels**: No significant blockages; normal.
**1 vessel**: One artery is blocked; mild disease.
**2 vessels**: Two arteries are blocked; moderate disease.
**3 vessels**: All three arteries are blocked; severe disease

As the number of affected vessels increases, so does the risk of heart complications and the need for more intensive treatment

# 4. Methodology

In this section, we will outline the methodology used in this data analysis, including the application of descriptive statistics and visualizations.

## A. Descriptive Statistics

To gain an initial understanding of the dataset, we began with the application of descriptive statistics. This involved calculating key summary statistics for the dataset, which consists of 946 rows and 13 variables. The following descriptive statistics were computed:

- **Data Distribution**: Histograms were generated to visualize the distribution of Age, providing insights into data skewness and kurtosis.
- **Measures of Central Tendency:** We calculated the mean, median, and mode for relevant variables to understand the central tendency of the data.
- **Measures of Dispersion:** We determined the range, standard deviation, and variance to assess the spread and variability of the data.

## B. Visualizations

Visualizations were a crucial component of our analysis, aiding in the interpretation of data patterns and trends. We employed a variety of data visualizations, including:

- **Box Plots:** Box plots were employed to identify outliers and understand the distribution of variables like 'classification of heart disease' within different categories, such as 'Resting BP' or 'Maximum Heart Rate' or 'Serum Cholesterol'.
- **Bar Charts:** We created bar charts to show the classification of heart disease and the risk factors in the dataset.
- **Histograms:** Histograms provided insights into the distribution of continuous variables, showing the frequency of data points within specific intervals.
- **Scatter Plots:** Scatter plots were used to visualize relationships between two continuous variables, such as 'age' and 'serum cholesterol' helping us identify potential correlations.
- **Pie chart:** we create pie charts to show the gender variable.

These visualization techniques enhanced our ability to identify patterns, outliers, and potential relationships within the dataset.

By combining descriptive statistics and visualizations, we aimed to provide a comprehensive analysis of the dataset, revealing valuable insights into the factors influencing heart diseases. According to that we expect to outline relationships based on the below categories;

**1.Relationships between numerical readings taken about heart and the tendency to have heart diseases**

Our goal here is to identify ranges of readings present in patients who have heart diseases and who don't moreover we hope to identify whether a higher mean reading leads to higher or lower chances of having heart diseases

**2. Relationships between heart conditions categorised by order of severity and presence or absence of heart diseases**

Some of the heart readings and conditions have been arranged by degree of severity (good, better, worse). Our goal is to identify how the degree of severity of the readings/ conditions affect the presence or absence of heart diseases

**3. Relationship between blood vessel characteristics and presence of absence of heart diseases**

Variables such as slope, no of major vessels... deal with characteristics of blood vessel. Our goal is to identify how these characteristics affect absence or presence of heart diseases.

**4. Relationship between different heart conditions**

To see if there are any relationships between different heart conditions

# 5. Objectives

### Project Objectives

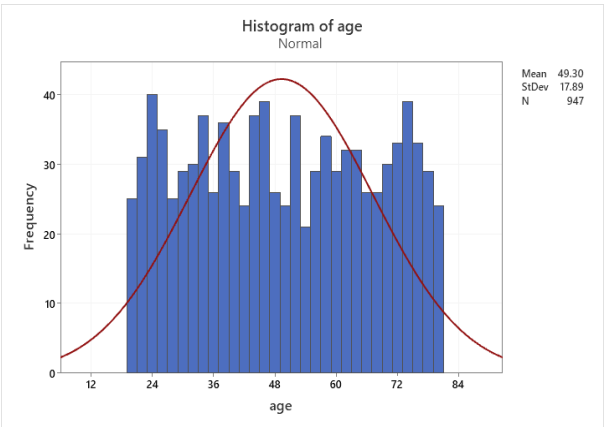
- **Data Understanding**: We aim to gain a deep understanding of the dataset, including the various attributes and their significance.
- **Data Exploration**: Our objective is to thoroughly explore the dataset, identifying key features, patterns, and trends.
- **Heart disease Determinants**: to investigate relationship between heart conditions and the presence or absence of heart diseases. We plan to understand which the specification of the heart condition tend to generate more heart disease.
- **Relationships between heart conditions** : to observe patterns between different conditions
- **Impact Assessment**: We intend to quantify the impact of various factors on used heart diseases to understand how specific attributes influence presence or absence of heart diseases.
- **Data Visualization**: We will use visual tools like charts and graphs to effectively represent the data, highlighting patterns and insights that may be less obvious in raw form.
- **Statistical Analysis**: We will apply statistical methods to analyze the dataset, employing both descriptive statistics to summarize the data and inferential statistics to draw informed conclusions.

In essence, our objectives focus on understanding the dataset, uncovering insights, and offering practical advice for understanding the nature of heart diseases.

# 6. General analysis

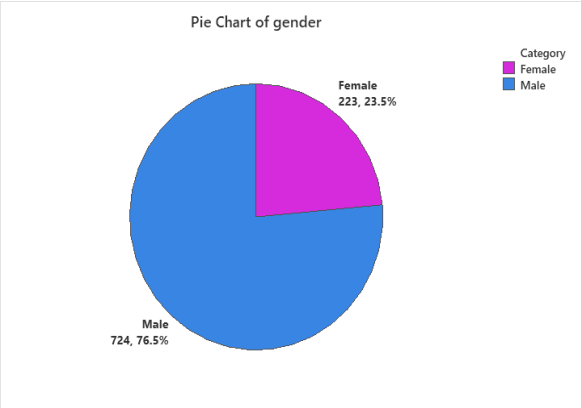
## i. Age

### Histogram 1




## ii. Gender

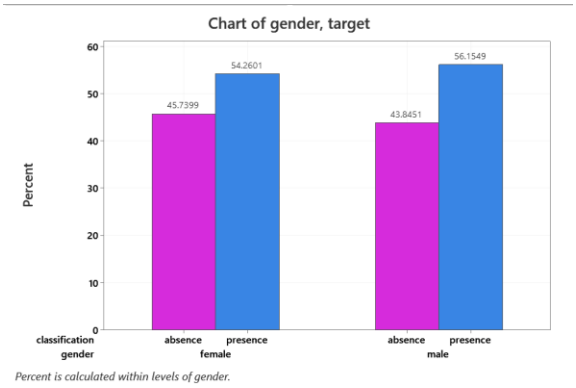
### Pie chart 1



Above histogram shows the ages of 946 patients. And the average of patients is 49 while the most detecting age group is 20 to 40. Overall, the ages are quite evenly distributed and reflect an accurate cross section of society as all the different age groups are present without any extreme peaks between 20 and 80.

This is a pie chart that shows the gender distribution of our data-set. It consists of 76.5% of male patients and 23.5% of female patients. So, we can say most of male patients tend to do heart related check-ups more than women. In other words they assume they are at a higher risk of getting heart diseases.

## iii. Gender vs Classification of heart disease



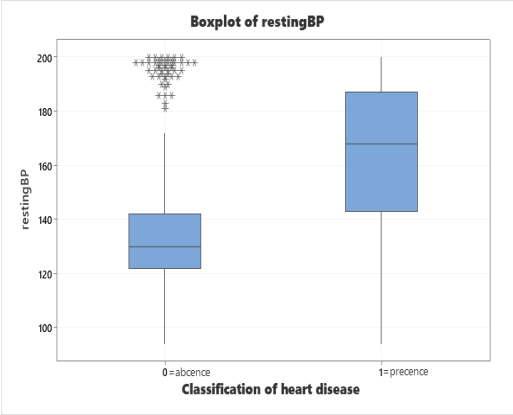

Cluster bar chart 1

This cluster bar chart shows the distribution of heart disease presence across male and female genders. There doesn't seem to be a big difference between presence and absence in male and females. A higher percentage of females have **heart disease (presence)** compared to those who do not have heart disease (**absence**). Similarly, a higher percentage of males have **heart disease (presence)** compared to those who do not have heart disease (**absence**). So, both **males** and **females** show a higher percentage in the **presence** category compared to the **absence** category, with the trend being more pronounced in males. But, overall the presence or absence of heart disease does not seriously depend on gender. Moreover it is hard to get a complete clear idea on the impact of gender on presence of disease because over 70% of the data set comprises of males.

# 7.Relationships between numerical readings taken about heart and the tendency to have heart diseases

## I. Boxplot for resting BP

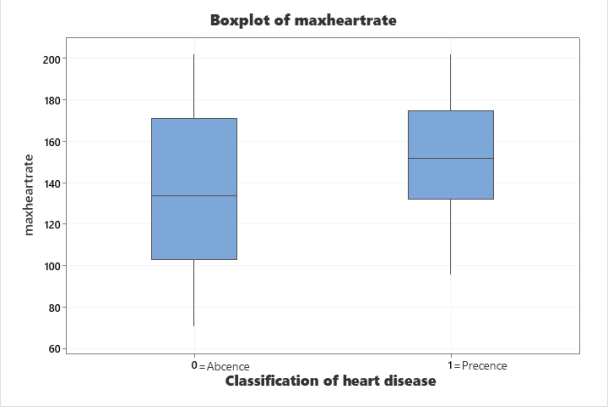
Box plot 1




This box plot represents the distribution of resting blood pressure when there is no heart disease or heart disease. Considering both two box plots the median resting BP's are 130mmHg and 170mmHg. So this clearly shows that individuals with heart disease have a higher median resting BP compared to those without heart disease. The presence of outliers without heart disease who still have high blood pressure could suggest the presence of an early risk of heart disease. Overall box plot reveals a clear difference in resting blood pressure levels between individuals with and without heart disease, and blood pressure is associated with a higher risk of heart disease.

## ii. Boxplot for maximum heart rate

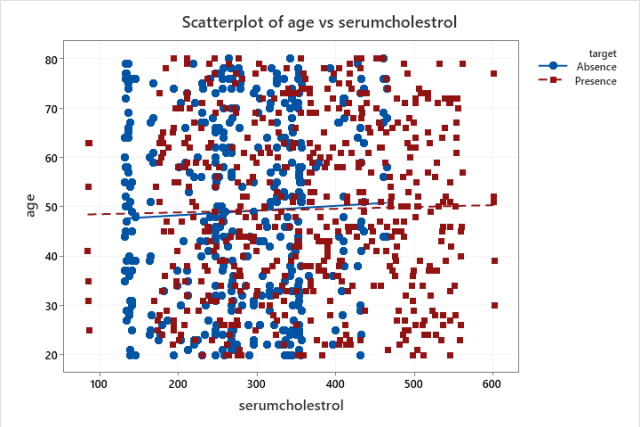
Box plot 3




The above boxplots show the maximum heart rates with and without heart disease. Considering both boxplots the medians are 150 bpm and 135 bpm, this difference suggests that, on average, individuals with heart disease achieve a lower maximum heart rate compared to those without heart disease. The group without heart disease has a wider IQR(inter quartile range) and the group with heart disease has a narrow IQR and shorter whisker.

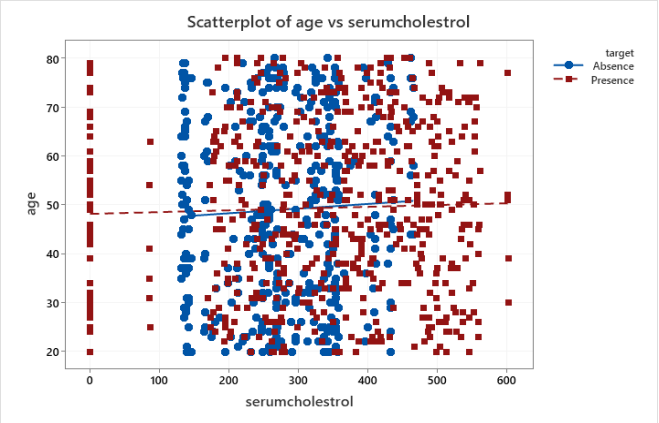
# iii. Scatterplot of age Vs Serum Cholesterol Vs Classification of heart disease.

Scatter plot 1




This scatter plot provides insight into how age and resting serum cholesterol may vary with the target. Based on the distribution of serum cholesterol levels spread across the full range of ages, it doesn't show a strong relationship between the above two variables. There is no special trend in the distribution of ages or cholesterol for this group. The dot point spread all over the chart but there seems to be a higher concentration of red dots at higher cholesterol levels compared to blue, special age in 40 to 60.The trend line is a slightly positive connection between age and cholesterol that means when age increases ,serum cholesterol tends to slightly increase. Considering the target variable the chart shows individuals with higher serum cholesterol levels have higher risk, especially in middle and older people

Scatter plot 2




**P.S.** In the original data set, there were 54 missing values in the serum cholesterol variable, therefore before we removed those values we analyzed them using this scatterplot, and after removing the values we got another scatterplot, considering both graphs there were no differences. So removing missing values does not affect the original conclusions.

## iV. Descriptive statistics

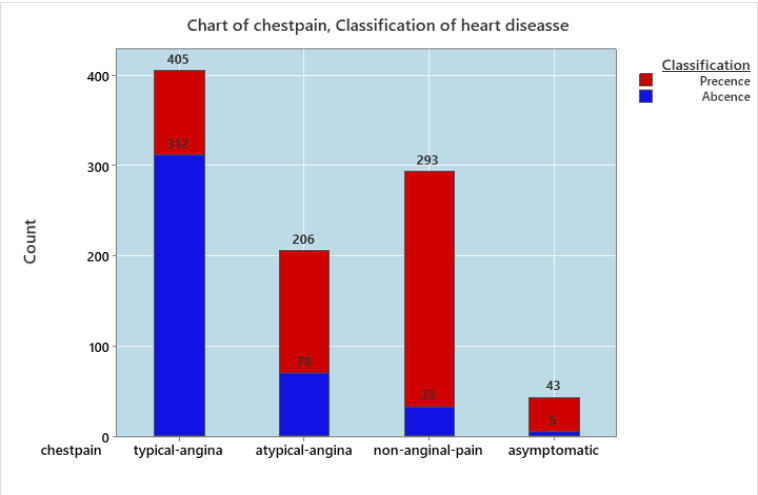
**Statistics**

| Variable | N | N* | Mean | StDev | Minimum | Median | Maximum |
|---|---|---|---|---|---|---|---|
| restingBP | 947 | 0 | 151.07 | 30.07 | 94.00 | 144.00 | 200.00 |
| serumcholestrol | 947 | 0 | 328.88 | 113.07 | 85.00 | 326.00 | 602.00 |
| maxheartrate | 947 | 0 | 145.01 | 34.48 | 71.00 | 145.00 | 202.00 |

This represents descriptive statistics for three variables considering them, which seems to have relatively symmetrical distributions around their median, there may not be extreme skewness in the data but the spread can give more insight into the concentration of data points around the mean.

# 8. Relationships between heart conditions categorised by order of severity and presence or absence of heart diseases

## i. Chest pain type vs Classification of heart disease




Stack bar chart 1

According to above stacked bar chart,

# typical angina chest pain type has the highest counts in both red and blue categories. Considering typical angina majority who have it don't seem to have heart disease (312 have heart diseases while 93 do have the disease

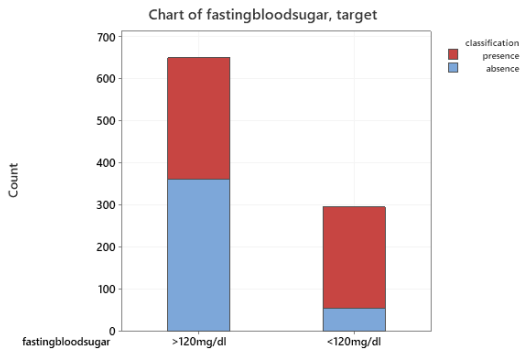
# for atypical angina majority who have this condition have heart diseases (136 have heart diseases 70 don't have it).

# for non-angina pain the vast majority (approximately 90%) of the who have this condition have heart diseases. (260 have heart diseases, 33 don't have it)

# for Asymptomatic pain a similar trend can be observed where the majority who have the condition have the disease. Overall, a clear distinction can be made with the type of chest pain and the chances of

having heart diseases. typical and atypical angina have roots which can be more easily detected (so medication can be taken) therefore they don't seem to show a majority of the presence of heart diseases while with non-angina and asymptomatic pain, detecting the root is harder (so medication cannot be taken) therefore they give out a majority of patients with heart diseases.

## ii. Fasting blood sugar VS Classification of heart disease

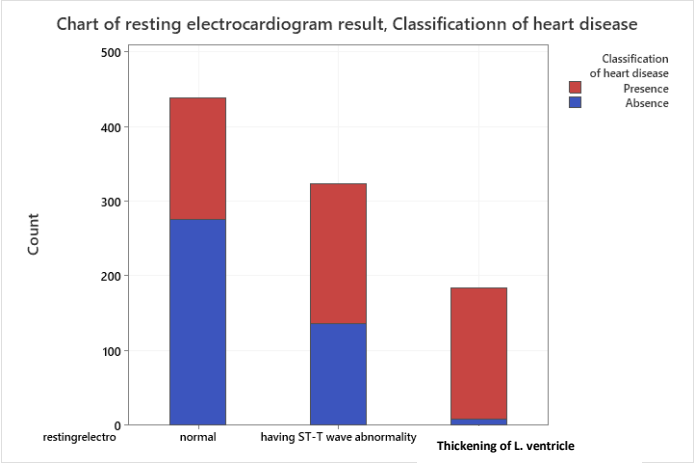



Stack bar chart 2

The above chart shows the relationship between fasting blood sugar levels and the presence or absence of heart diseases. Most individuals with blood sugar below 120 don't have the heart diseases. For those with blood sugar above 120, have more cases of heart diseases. This suggests that high blood sugar level is highly affects the heart disease.

## iii. Resting electrocardiogram VS Classification of heart disease

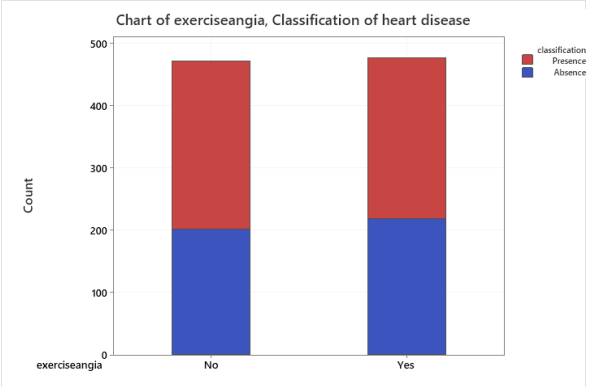
Stacked bar chart 4




The bar chart shows the distribution of a variable called "resting electrocardiogram" across two categories of a variable named "target", that represent having or not heart disease. Category labeled "Normal" category has the highest number of observations, with a roughly even split between the target and category, that labeled as 'showing probable' has the fewest observations overall, but nearly everyone has heart disease. This suggests the presence of heart disease, indicating that individuals with this value are much more likely to have heart disease

## iv. Exercise angina VS Classification of heart disease

Stack bar chart 2

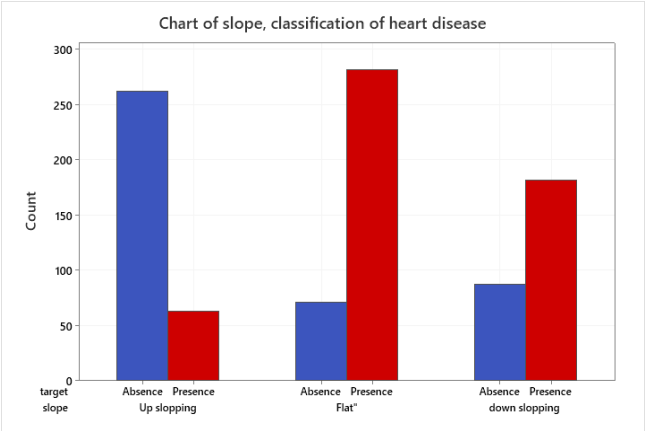



Examining the chart, we notice that the total count for each category of "exercise angina" is nearly 500. In the case of no exercise angina, approximately half of the count is assigned to presence of heart disease, while the remaining portion is associated with presence of heart disease. Likewise, for presence of exercise angina category, the distribution is almost the same as no exercise angina category, with approximately half of the total count falling into absence of heart disease and the other half into presence of heart disease. Overall, we can say that exercising angina has no impact on heart disease.

# 9. Relationship between blood vessel characteristics and presence of absence of heart diseases

## i. Slope VS Classification of heart disease

Cluster bar chart 2



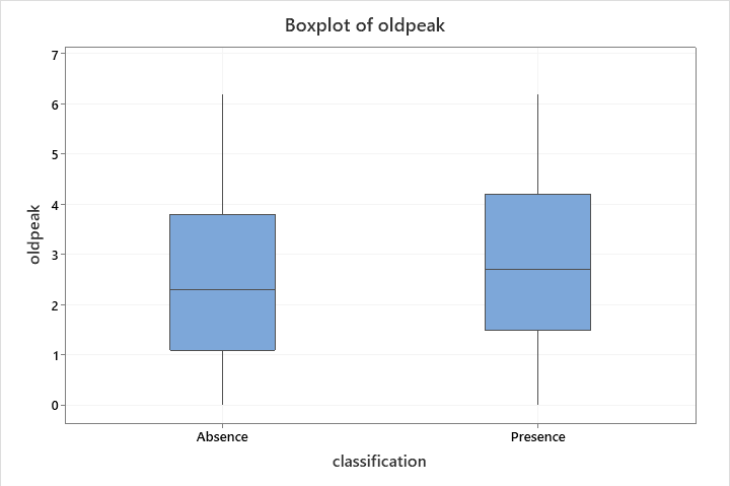

Based on the data regarding heart rate slope and the presence of heart disease, we can observe distinct patterns.

Individuals with an upsloping heart rate slope are less likely to have heart disease compared to those with flat or down sloping slopes. Among those with flat and down sloping heart rate slopes, a higher number have heart disease, with the flat slope group having the highest prevalence. The group with down sloping heart rate slopes shows the least number of individuals without heart disease.

This suggests that heart rate slope might significantly influence the likelihood of having heart disease, as seen by the varying outcomes across different slope categories. Overall, it appears that having a flat or heart rate down sloping heart rate slope is more strongly associated with heart disease compared to an upsloping slope.

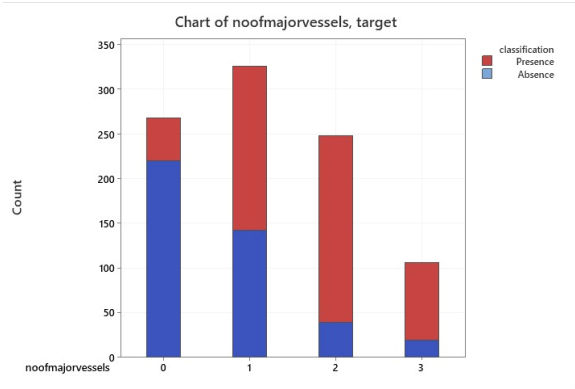
# iii. Old peak VS Classification of heart disease

Box plot 4




This graph displays the distribution of a variable called "old peak" for two different categories labeled as "Absence" and "Presence" under the "classification" axis. Both categories have similar median values, approximately at 2. This indicates that the central tendency of the "old peak" variable is nearly the same for both. In summary, individuals with heart disease have a more consistent distribution of "old peak" values, while those without heart disease show greater variability and a wider range of values.

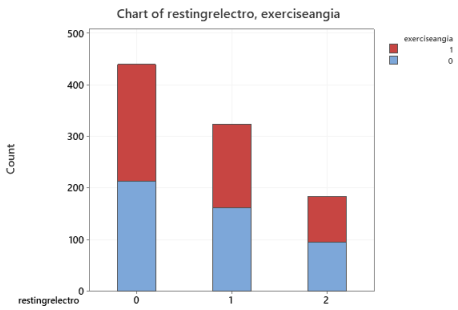
# iv.No of major vesseles vs Classification of heart desease.

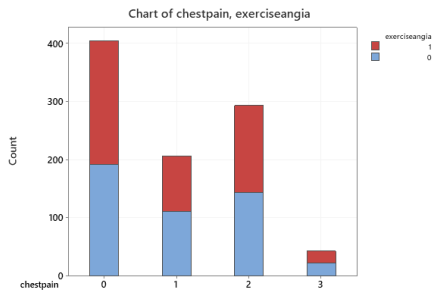


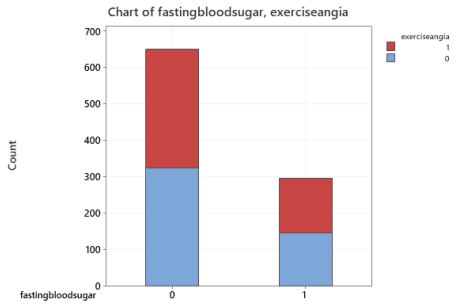

This graph shows the count of major vessels, across the heart disease classification. In 0 major vessels there is a slightly higher amount of absence than presence but other three major vessels (1,2,3) have presence significantly higher than absence. So, the count of 1, 2, 3 no of major vessels have high risk for having heart disease.

# 10. Relationship between different heart conditions

A meaningful analysis cannot be taken from observing relationships between heart conditions as shown by the graphs below. The reasons for this might be the fact that these medical conditions relate to different aspects and issues of the heart which are independent from each other.




# 11. Conclusion

In conclusion the data set has provided us with a comprehensive understanding of the diagnostics and their specific conditions which lead to a higher chance of having heart diseases. (The reasons for this might be the fact that these medical conditions relate to different aspects and issues of the heart which are independent from each other.)

A major inference that can be made when studying this data set is that a meaningful analysis can mostly be taken only when comparing the heart conditions and the presence or absence of heart diseases.

For heart conditions taken in the form of reading the data set showed us if the value provided by these medical tests were higher they lead to higher chances of heart diseases.

Moreover, heart conditions were separated based on severity (chest pain types, slopes) and how severe the attribute, the higher the chance of getting heart diseases.

The data consisted of a wide range of ages so it provided us an overall insight on these conditions affect risk of heart diseases irrespective of age.

The data set was majoritively composed of male patients so it is hard to get a clear understanding on how the risk of getting heart diseases affect males and females separately.

## Key findings

**Male** patients tend to do heart related check-ups more than **females**

**Chest pain type -** typical and atypical angina have roots which can be more easily detected( so medication can be taken) therefore they don't seem to show a majority of the presence of heart diseases while with non-angina and asymptomatic pain, detecting the root is harder (so medication cannot be taken) therefore they give out a majority of patients with heart diseases.

The mean given for **resting blood pressure, serum cholesterol, maximum heart rate, old peak** give higher values for presence of heart diseases than absence of heart diseases

**Resting electrocardiogram results** - when they show deviations from the normal the chances of heart diseases increase

**Slope** when it is flat it shows the highest risk of heart diseases while down sloping shows second highest risk and flat shows least like risk

When the **number of major vessels** blocked increases the chances for heart diseases increases

**Exercise angina** which acts up when exercising doesn't have an impact on presence on absence of heart diseases

Overall, our analysis reveals that while no single factor alone determines heart disease risk, the interplay between age, type of chest pain, gender, and resting ECG values provides a comprehensive picture of an the risk of disease varies with different conditions. These insights can guide more effective screening and diagnostic strategies, ultimately contributing to better preventive and therapeutic measures for heart disease.

**Comparison with real world values**

The Median value obtained from the box plot drawn for blood pressure of patients with presence of absence of diseases, shows us that the patients without heart diseases have a median of 130mmHg. The medically recommended healthy blood pressure for men of age 40-56 is 124 mm Hg and that of women is 122 mm Hg.

A fasting blood sugar level less than 100 mg/dL (5.6 mmol/L) is normal. Through the graphs we obtained comparing fasting blood sugar and presence or absence of heart diseases, it was found that when the fasting blood sugar is less than 120mg/dl the chances of getting heart diseases are less.

In general we can say that the medically recommended values for heart conditions tally with the means or the majority of the patients without heart diseases. We can roughly assume that the data set is an accurate description of society.

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